

Q1.

The time in hours to failure of a component may be modelled by an exponential distribution with parameter $\lambda = 0.025$
 In a manufacturing process, the machine involved uses one of these components continuously until it fails.

The component is then immediately replaced.

- (a) Write down the mean time to failure for a component. (1)
- (b) Find the probability that a component will fail during a 12-hour shift. (1)
- (c) A component has not failed for 30 hours. Find the probability that this component lasts for at least another 30 hours. (2)
- (d) Find the probability that a component does **not** fail during 4 consecutive 12-hour shifts. (3)
- (e) (i) State the distribution that can be used to model the number of components that fail during one hour of the manufacturing process. (2)
- (ii) Hence, or otherwise, find the probability that no components fail during 5 consecutive 12-hour shifts. (2)

(Total 11 marks)

Q2.

- (a) The continuous random variable X follows an exponential distribution if it has a probability density function

$$f(x) = \begin{cases} ke^{-kx} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

where k is a positive constant.

- (i) Prove that the mean value, μ , of X is $\frac{1}{k}$. (3)
- (ii) Find, in terms of k , the median value, m , of X and hence show that $m < \mu$. (5)
- (b) The number of radioactive particles striking a screen in a time period of length t seconds follows a Poisson distribution with mean $\frac{t}{\lambda}$, where λ is a constant.

- (i) Write down the probability that no particles strike the screen in a period of t seconds.

(1)

- (ii) The random variable T is defined as the length of time, in seconds, between successive radioactive particles striking the screen.

(A) Show that

$$P(T < t) = 1 - e^{-\frac{t}{\lambda}}$$

(2)

- (B) Hence, by finding the probability density function of T , state the distribution of T .

(2)

(Total 13 marks)

Q3.

The time, D days, between successive accidents at a factory can be modelled by an exponential distribution with mean 16.

- (a) Write down the numerical value for the standard deviation of D .

(1)

- (b) Calculate the probability that the time between successive accidents at the factory is

(i) more than 20 days;

(ii) between 20 and 30 days.

(5)

- (c) Given that there are no accidents during a 20-day period, determine the probability that there are no accidents during the next 20 days. Justify your answer.

(3)

- (d) Given that the factory is open five days each week, specify the distribution of the **weekly number** of accidents at the factory.

(2)

(Total 11 marks)